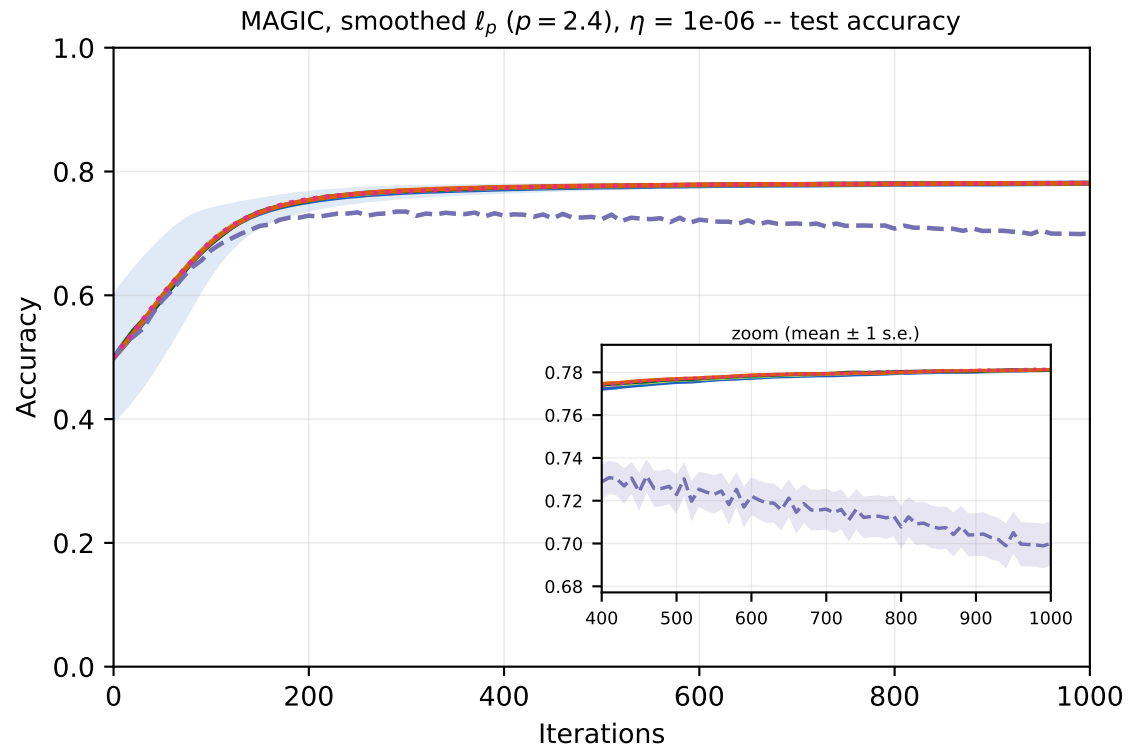
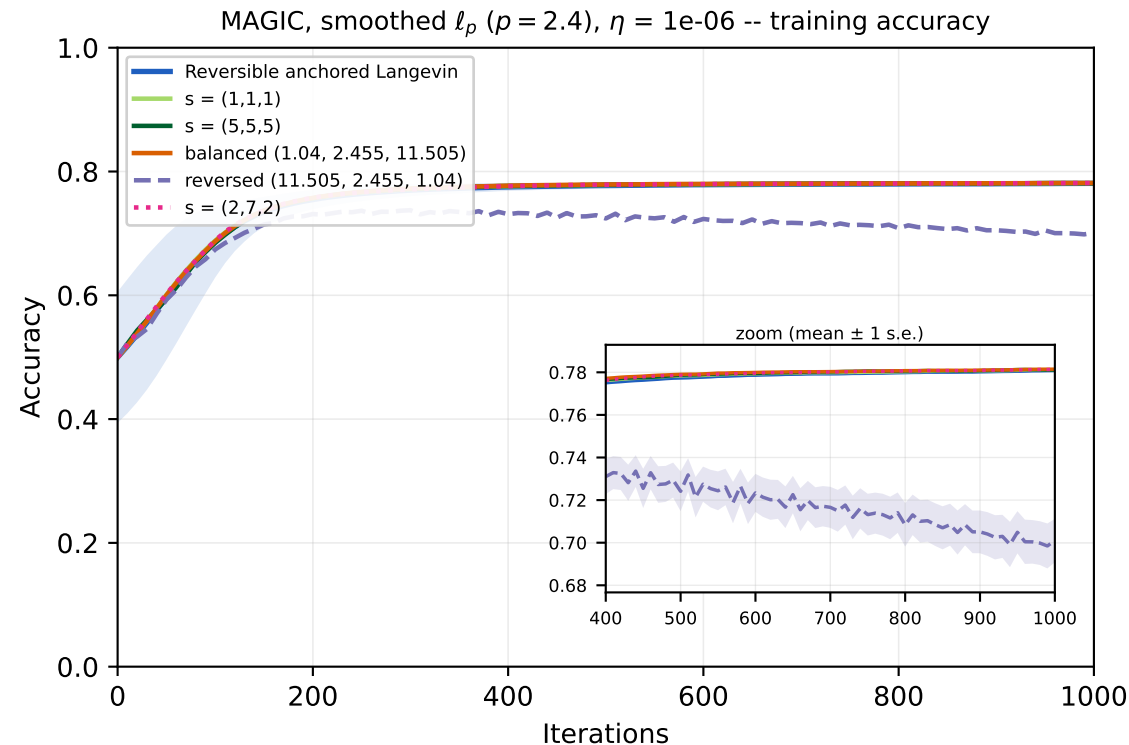


magic_lp: per-block strengths at $\eta = 1\text{e-}06$ (mix)



MAGIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.2$, $\text{Lambda} = 4$. $d = 10$ (intercept $w_0 + 9$ features). $U = \sum \text{softplus} - y \cdot x \cdot w + w_0^2 / (2 \sigma^2) + \text{lambda} \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \text{delta}^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$, $\text{lambda}_{\text{lasso}} = 152 = 0.01 \cdot n_{\text{train}}$, $\text{delta} = 0.000506$ (a in $[0.50, 1]$). Step $\eta = 1\text{e-}06$. Per-block strengths $s = (s_1, s_2, s_3)$ on $I_1 = \text{PC1}..\text{PC3}$, $I_2 = \text{PC4}..\text{PC6}$, $I_3 = \text{PC7}..\text{PC9}$. Isotropic vs gradient-balanced vs reversed-balanced vs hand-picked. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Band: mean $\pm$ 1 SD (ddof = 1) for the reversible arm -- repeat-run variability at a fixed split, not a confidence or credible interval. All triples are reported in results/blocks/blocks.csv; nothing here was selected.